

Linear Regression: Mathematical Notes

June 14, 2026

Dataset: Advertising Spend vs. Sales

Suppose a small company tracked its weekly advertising spending (in thousands of dollars) and the corresponding weekly sales (in thousands of dollars). We want to understand the relationship and predict sales based on advertising spend.

Week	Advertising Spend (x , in \$1000)	Sales (y , in \$1000)
1	1	3
2	2	5
3	3	6
4	4	8

Table 1: Advertising Spend vs. Sales Dataset

Goal: Find the best-fit linear model $y = mx + c$ that predicts Sales (y) based on Advertising Spend (x).

I. Ordinary Least Squares (OLS) Linear Regression

Linear Regression models the relationship between a dependent variable (y) and an independent variable (x) by fitting a linear equation to observed data. In Ordinary Least Squares (OLS) we find the line that minimizes the sum of the squared differences between the observed and predicted values.

Model Equation

$$y = mx + c$$

where:

- y is the dependent variable (output).
- x is the independent variable (input).

- m is the **slope** — the change in y for a one-unit increase in x .
- c is the **y-intercept** — the predicted value of y when $x = 0$.

The predicted value for observation i is $\hat{y}_i = mx_i + c$.

Objective Function (Least Squares)

We minimize the Residual Sum of Squares (RSS):

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (mx_i + c))^2$$

Normal Equations

Taking partial derivatives of RSS with respect to m and c , setting them to zero, and solving yields the closed-form solutions:

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad c = \bar{y} - m\bar{x}$$

where $\bar{x}$ is the mean of all x_i values and $\bar{y}$ is the mean of all y_i values.

Calculations for Simple Linear Regression

Step 1: Compute Sums and Means

Number of data points: $n = 4$.

Week	x_i	y_i	x_i^2	y_i^2	$x_i y_i$
1	1	3	1	9	3
2	2	5	4	25	10
3	3	6	9	36	18
4	4	8	16	64	32
Sum	$\sum x_i = 10$	$\sum y_i = 22$	$\sum x_i^2 = 30$	$\sum y_i^2 = 134$	$\sum x_i y_i = 63$

Table 2: Summary of computed values

Calculate the means:

$$\bar{x} = \frac{\sum x_i}{n} = \frac{10}{4} = 2.5 \qquad \bar{y} = \frac{\sum y_i}{n} = \frac{22}{4} = 5.5$$

Step 2: Compute Deviation Products

x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
1	3	-1.5	-2.5	3.75	2.25
2	5	-0.5	-0.5	0.25	0.25
3	6	0.5	0.5	0.25	0.25
4	8	1.5	2.5	3.75	2.25
Sum				8.00	5.00

Table 3: Deviation products used in the SS formula

Step 3: Calculate m (slope)

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8.00}{5.00} = 1.6$$

Step 4: Calculate c (y-intercept)

$$c = \bar{y} - m\bar{x} = 5.5 - (1.6)(2.5) = 5.5 - 4.0 = 1.5$$

Step 5: Regression Equation

The best-fit linear model is:

$$\hat{y} = 1.6x + 1.5$$

Step 6: Make a Prediction

If advertising spend is \$5,000 (i.e. $x = 5$):

$$\hat{y} = 1.6(5) + 1.5 = 8.0 + 1.5 = 9.5$$

So if the company spends \$5,000 on advertising, the model predicts \$9,500 in sales.

Dataset: Years of Experience vs. Salary

A company collected data on employees' years of experience and their corresponding annual salary (in thousands of dollars). We want to build a linear model to **predict salary** based on years of experience.

Employee	Experience (x , years)	Salary (y , in \$1000)
1	3	30
2	8	57
3	9	64
4	13	72
5	3	36
6	6	43
7	11	59
8	21	90
9	1	20
10	16	83

Table 4: Years of Experience vs. Salary Dataset

Goal: Find the best-fit linear model $y = mx + c$ that predicts Salary (y) based on Years of Experience (x), and use it for prediction.

I. Ordinary Least Squares (OLS) Linear Regression

Model Equation

$$y = mx + c$$

where m is the **slope** and c is the **y-intercept**.

Normal Equations (Closed-Form Solution)

$$m = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad c = \bar{y} - m\bar{x}$$

II. Calculations

Step 1: Compute Means

Number of data points: $n = 10$.

$$\bar{x} = \frac{\sum x_i}{n} = \frac{3 + 8 + 9 + 13 + 3 + 6 + 11 + 21 + 1 + 16}{10} = \frac{91}{10} = 9.1$$

$$\bar{y} = \frac{\sum y_i}{n} = \frac{30 + 57 + 64 + 72 + 36 + 43 + 59 + 90 + 20 + 83}{10} = \frac{554}{10} = 55.4$$

Step 2: Compute Deviation Products

x_i	y_i	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$
3	30	$3 - 9.1 = -6.1$	$30 - 55.4 = -25.4$	154.94	37.21
8	57	$8 - 9.1 = -1.1$	$57 - 55.4 = 1.6$	-1.76	1.21
9	64	$9 - 9.1 = -0.1$	$64 - 55.4 = 8.6$	-0.86	0.01
13	72	$13 - 9.1 = 3.9$	$72 - 55.4 = 16.6$	64.74	15.21
3	36	$3 - 9.1 = -6.1$	$36 - 55.4 = -19.4$	118.34	37.21
6	43	$6 - 9.1 = -3.1$	$43 - 55.4 = -12.4$	38.44	9.61
11	59	$11 - 9.1 = 1.9$	$59 - 55.4 = 3.6$	6.84	3.61
21	90	$21 - 9.1 = 11.9$	$90 - 55.4 = 34.6$	411.74	141.61
1	20	$1 - 9.1 = -8.1$	$20 - 55.4 = -35.4$	286.74	65.61
16	83	$16 - 9.1 = 6.9$	$83 - 55.4 = 27.6$	190.44	47.61
Sum				1269.60	358.90

Table 5: Deviation products used in the OLS formula

Step 3: Calculate m (Slope)

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{1269.60}{358.90} \approx 3.538$$

Step 4: Calculate c (y-Intercept)

$$c = \bar{y} - m\bar{x} = 55.4 - (3.538)(9.1) = 55.4 - 32.196 \approx 23.2$$

Step 5: Regression Equation

The best-fit linear model is:

$$\hat{y} = 3.538x + 23.2$$

Step 6: Make a Prediction

Question: Predict the salary of an employee with **12 years** of experience.

$$\hat{y} = 3.538(12) + 23.2 = 42.456 + 23.2 = 65.656$$

So the model predicts a salary of approximately **\$65,656** for an employee with 12 years of experience.